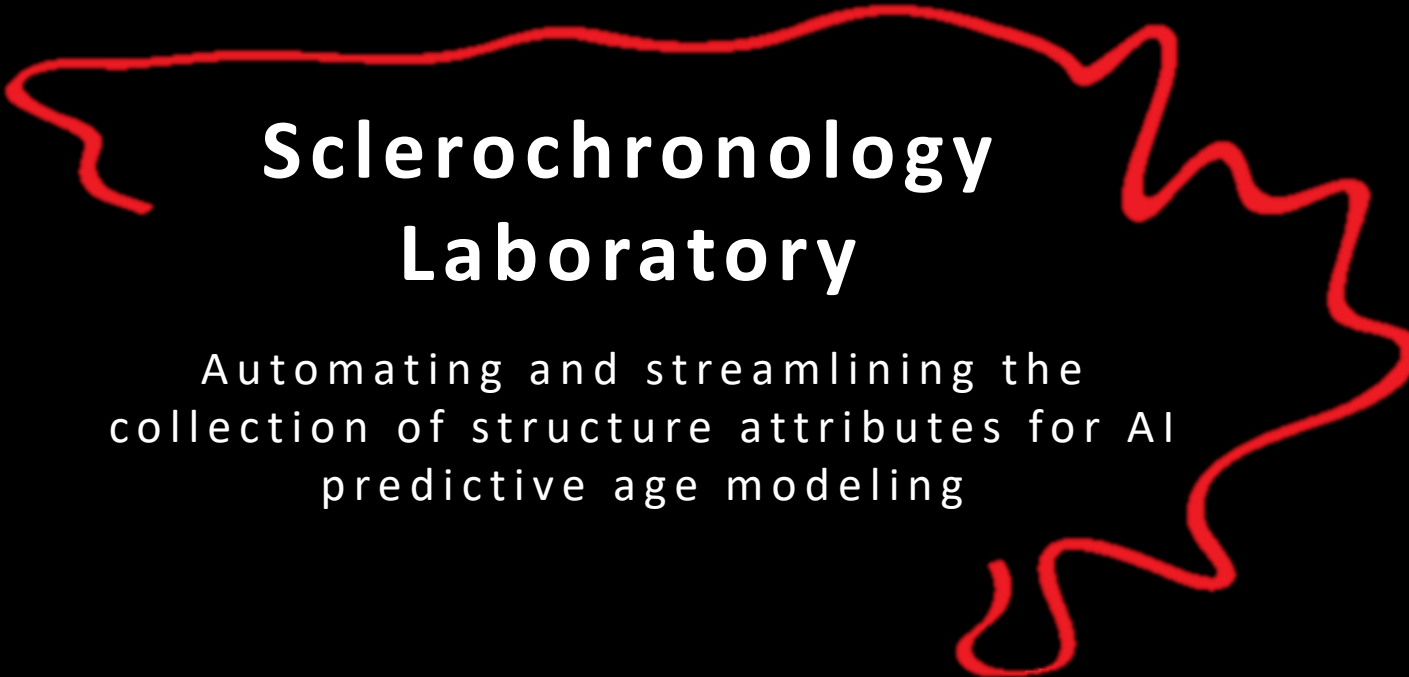


A thick, hand-drawn style red line that forms a horizontal, wavy shape, resembling a stylized wave or a brushstroke. It is positioned behind the text, with the text appearing as if it is floating within or attached to the line.

Sclerochronology Laboratory

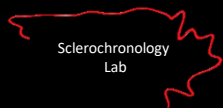
Fisheries & Oceans Canada
Pacific Biological Station

A thick, hand-drawn red line that forms a horizontal, wavy shape, resembling a stylized cloud or a brushstroke, framing the text.

Sclerochronology Laboratory

Automating and streamlining the
collection of structure attributes for AI
predictive age modeling

Presented by Stephen Wischniowski and Sarah Gravel



Sclerochronology Lab Research Branch Alternative Age Estimation

Human produced age estimates has become a bottleneck for fisheries management. Alternative means of estimating fish age is generally targeting;

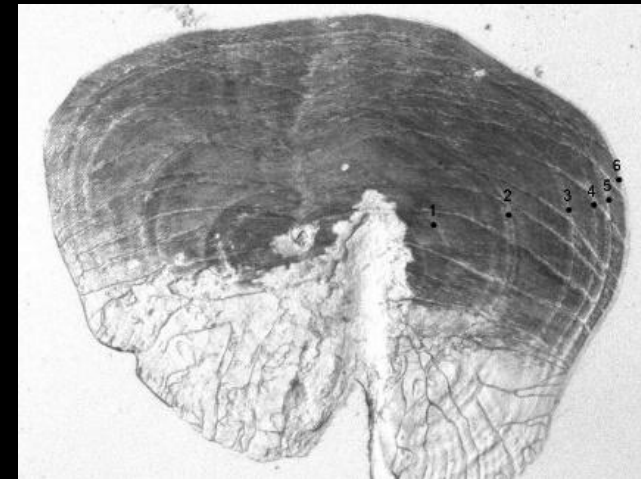
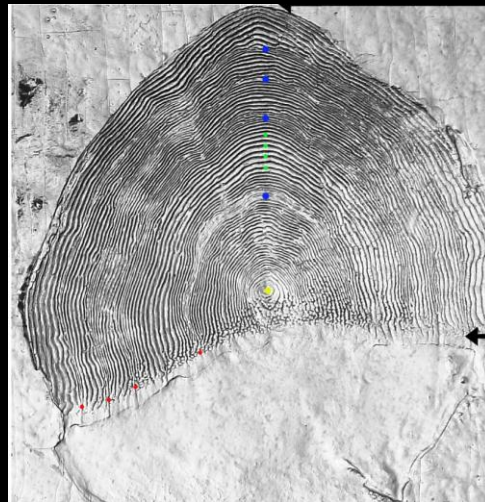
- high-volume age requests,
- reducing turnover times, increasing output and productivity,
- addressing budget cuts and departures of skilled personnel, and
- freeing up research time for less the frequently requested and more difficult to age species.

This, however, will only shift the ageing bottleneck to the collection and processing of structure attributes required to feed predictive age models. As Alternative Age Estimation (AAE) advances, more capacity will result in increased demand for more ages.

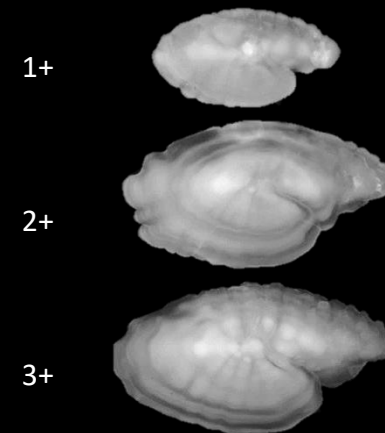
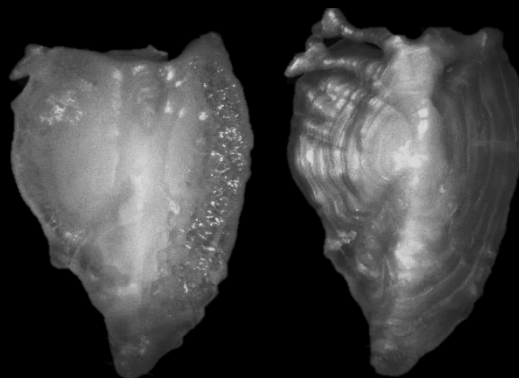
Creating an automated and streamlined throughput for structure attribute collection will increase the effectiveness and process output to feed AI models.

Alternative Age Estimation Structures

Structures:
Scale - salmon and herring



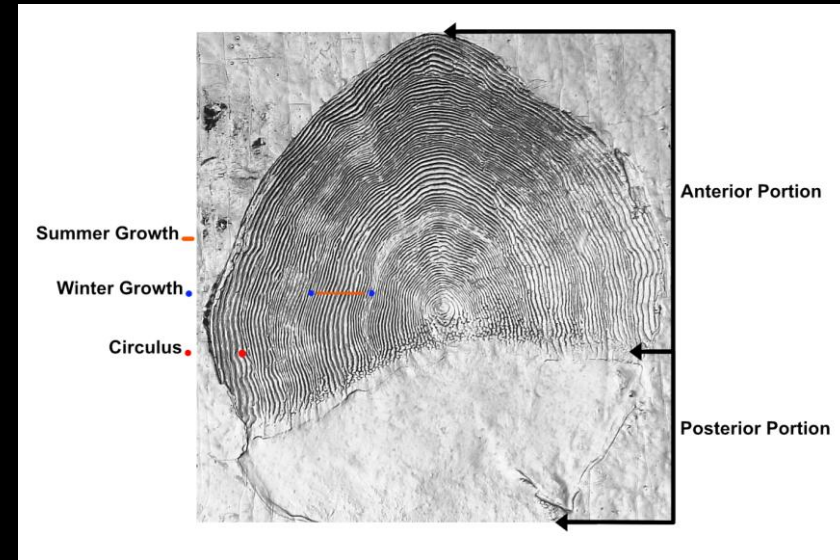
Otolith – groundfish



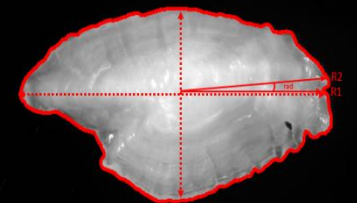
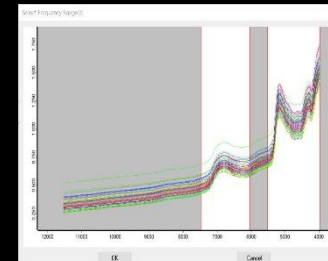
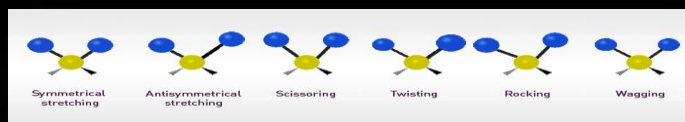
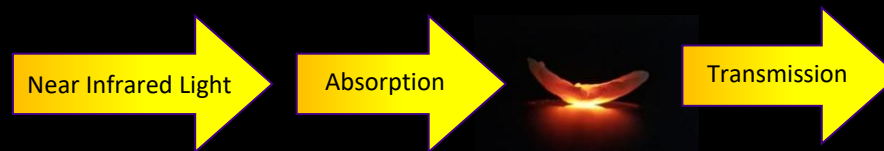
Alternative Age Estimation Attributes

Structure Attributes:

Scale -growth pattern



Otolith -Infrared (IR) scans, weight and shape

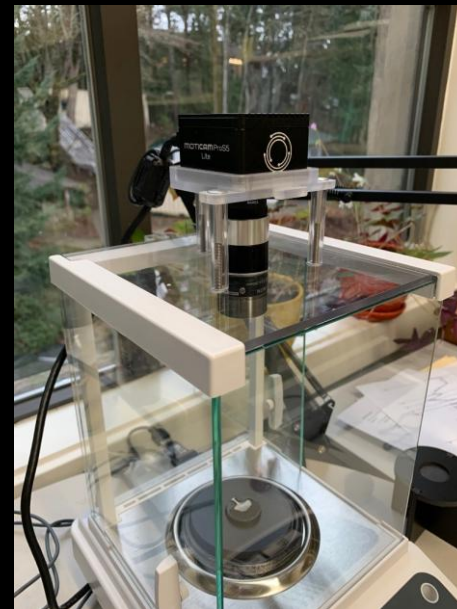
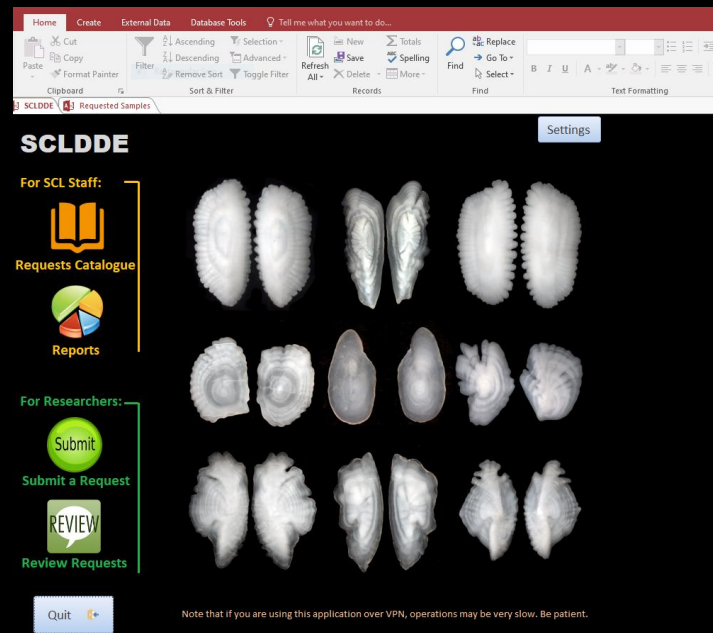


Alternative Age Estimation Equipment



Method of groundfish otolith age and attribute data collection

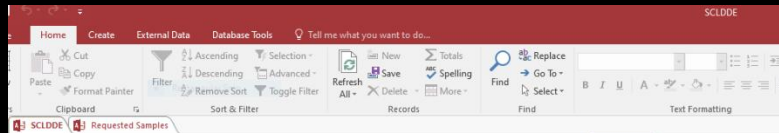
- Sclerochronology Lab Direct Data Entry (SCLDDE) application
- Combination four decimal place balance and otolith surface image capture
- Near Infrared otolith spectroscopy



Alternative Age Estimation Equipment

Method of groundfish otolith age and attribute data collection

- Sclerochronology Lab Direct Data Entry (SCLDDE) application



SCLDDE

For SCL Staff:



Requests Catalogue



Reports

For Researchers:



Submit a Request



Review Requests

Quit



Note that if you are using this application over VPN, operations may be very slow. Be patient.

PBS Ageing Request

Species: Arrowtooth Flounder | Start: 2019 | End: 2020 | Size (cm): 0 to 500 (not required)

Sexes: ALL SEXES | Trip Type: ALL TRIP TYPES | Gear: ALL GEAR TYPES

NOT LOOKED AT | OBSERVED DOMESTIC | BOTTOM TRAWL
 MALE | CHARTER | MIDWATER TRAWL
 FEMALE | RESEARCH | LONGLINE
 LOOKED AT BUT UNKNOWN | NON - OBSERVED DOMESTIC | SHRIMP TRAWL
 OBSERVED J-V | TRAP
 POLISH COMM NATIONAL | HANDLINE
 RUSSIAN COMMERCIAL NA | UNKNOWN TRAWL
 POLISH COMMERCIAL SUP | JIG
 JAPANESE COMMERCIAL | GILLNET

Include Previously Aged Samples ☐

Run Close

Sample Id	Year	Activity	Trip Type	Capture Date	Vessel	Area	Set	Sample Date	Box Id	Tray Id	First Fish Num	Last Fish Num	Num Ages	Num Aged	Num To Age	Age Every Nth	Priority
?	2019	Hecate Strait Synoptic Bottom Trawl Survey	Charter	05/21/2019	Nordic Pearl	5C	24	05/21/2019	FF150	2019-011-012	343	385	40	0			
?	2019	Hecate Strait Synoptic Bottom Trawl Survey	Charter	05/21/2019	Nordic Pearl	5C	27	05/21/2019	FF150	2019-011-032	434	478	45	0			
?	2019	Hecate Strait Synoptic Bottom Trawl Survey	Charter	05/25/2019	Nordic Pearl	5D	42	05/25/2019	FF150	2019-011-032	510	531	22	0			
?	2019	Hecate Strait Synoptic Bottom Trawl Survey	Charter	05/25/2019	Nordic Pearl	5D	45	05/25/2019	FF150	2019-011-047	561	587	27	0			
?	2019	Hecate Strait Synoptic Bottom Trawl Survey	Charter	05/26/2019	Nordic Pearl	5D	54	05/26/2019	FF150	2019-011-043	777	803	27	0	10	2	3
?	2019	Hecate Strait Synoptic Bottom Trawl Survey	Charter	05/26/2019	Nordic Pearl	5D	61	05/26/2019	FF150	2019-011-043	862	887	26	0			
?	2019	Hecate Strait Synoptic Bottom Trawl Survey	Charter	05/28/2019	Nordic Pearl	5D	76	05/28/2019	FF150	2019-011-034	1084	1109	26	0			
?	2019	Hecate Strait Synoptic Bottom Trawl Survey	Charter	05/28/2019	Nordic Pearl	5D	80	05/28/2019	FF150	2019-011-034	1189	1216	28	0			
?	2019	Hecate Strait Synoptic Bottom Trawl Survey	Charter	05/29/2019	Nordic Pearl	5C	86	05/29/2019	FF150	2019-011-034	1364	1391	27	0			

Number of requested samples: 1
 Number of requested ages: 10

Submit Request Close

Alternative Age Estimation Equipment

Method of groundfish otolith age and attribute data collection

- Sclerochronology Lab Direct Data Entry (SCLDDE) application

Requests Catalogue

Request ID: Sample ID: Requester: Status: ☐ Not Started ☐ Completed ☐ In Progress ☐ Precision? ☐ Delete? ☐ Unresolved?

Species: Pacific Hake Vessel: Nordic Pearl Requester: ENT\GrandinC Assigned To: Mary-Jane Hudson

Submitted On: 10/20/2021 Due On: Started On: 11/9/2021 Completed On: 11/15/2021 Capture Date: 7/10/2021

Sample ID: 554923 Box ID: Tray ID: 2021-026-071 Num. Ages: 23 # Ages Requested: 23 Priority:

Fish # 1 to 23 Training? ☐ Comment: Wet weight recorded. Ageing Precision

Request ID: 100012 UR? ☐ Current Status: Completed (Processed) Precision? ☒ n = 2 Total = 70

Species: 225 Pacific Hake Vessel: Nordic Pearl Requester: ENT\GrandinC Assigned To: Mary-Jane Hudson

Submitted On: 10/20/2021 Due On: Started On: 11/9/2021 Completed On: 11/17/2021 Capture Date: 7/10/2021

Sample ID: 554926 Box ID: Tray ID: 2021-026-071 Num. Ages: 14 # Ages Requested: 14 Priority:

Fish # 24 to 37 Training? ☐ Comment: Wet weight recorded. Ageing Precision

Request ID: 100012 UR? ☐ Current Status: Completed (Processed) Precision? ☒ n = 4 Total = 70

Species: 225 Pacific Hake Vessel: Nordic Pearl Requester: ENT\GrandinC Assigned To: Mary-Jane Hudson

Submitted On: 10/20/2021 Due On: Started On: 11/10/2021 Completed On: 11/17/2021 Capture Date: 7/10/2021

Sample ID: 554935 Box ID: Tray ID: 2021-026-071 Num. Ages: 25 # Ages Requested: 25 Priority:

Fish # 38 to 62 Training? ☐ Comment: Wet weight recorded. Ageing Precision

Request ID: 100012 UR? ☐ Current Status: Completed (Processed) Precision? ☒ n = 5 Total = 70

Species: 225 Pacific Hake Vessel: Nordic Pearl Requester: ENT\GrandinC Assigned To: Mackenzie Beck

Submitted On: 10/20/2021 Due On: Started On: 11/15/2021 Completed On: 11/26/2021 Capture Date: 7/10/2021

Sample ID: 554947 Box ID: Tray ID: 2021-026-071 Num. Ages: 27 # Ages Requested: 27 Priority:

Fish # 64 to 90 Training? ☐ Comment: Ageing Precision

Groundfish Age Data Sheet

Species: Pacific Hake Vessel Name: Nordic Pearl Fish Num. Range: 1 to 23 Started On: 10/27/2021

Sample ID: 554923 Date Caught: 7/9/2021 Subsample: 23 of 23 Completed On: 11/19/2021

Area: 5A Set Number: 8 Storage Box: Tray Number: 2021-026-071

Sample Comment: Primary Precision Test Secondary Secondary

Type: Primary Reading Comment:

Method: Otolith Broken And Burnt

Date: 10/27/2021

Reader: Mary-Jane Hudson Effort: t= QCt= QCn=

Tray	Oto	Fish #	Wt (g)	Pic?	Flagged	# Cryst.	General Comment	PI	Class	Min	Max	EG	UC	Reader Comment
A01	1	0.2849							8	7	8	1		Check with 4th.
A02	2	0.2480							9	9	10	1		Vague 2nd
A03	3	0.2638							5	5	5	1		
A04	4	0.2756		Flagged			MJ Review, mixed fish		6	4	6	1		Vague 1st, ch with 3rd or vague 3rd.
A05	5	0.3064							9	8	9	1		poor burns - Vague
A06	6	0.3423							13	13	13	1		
A07	7	0.2894							9	9	9	1		chs with 2, 3 & 4th yrs.
A08	8	0.2760							10	9	10	1		off centered break.
A09	9	0.2448		Flagged			MJ Review		7	7	10	1		poor burns, vague, checks
A10	10	0.3369							11	11	11	1		
A11	11	0.3172							10	9	10	1		
A12	12	0.2674							12	11	12	1		
A13	13	0.2452							7	7	8	1		vague 1st, strong nucleus
A14	14	0.2587							10	9	11	1		vague 2nd
A15	15	0.2860							12	12	12	1		off centred break
A16	16	0.2959							12	12	12	1		
A17	17	0.2857							10	8	10	1		

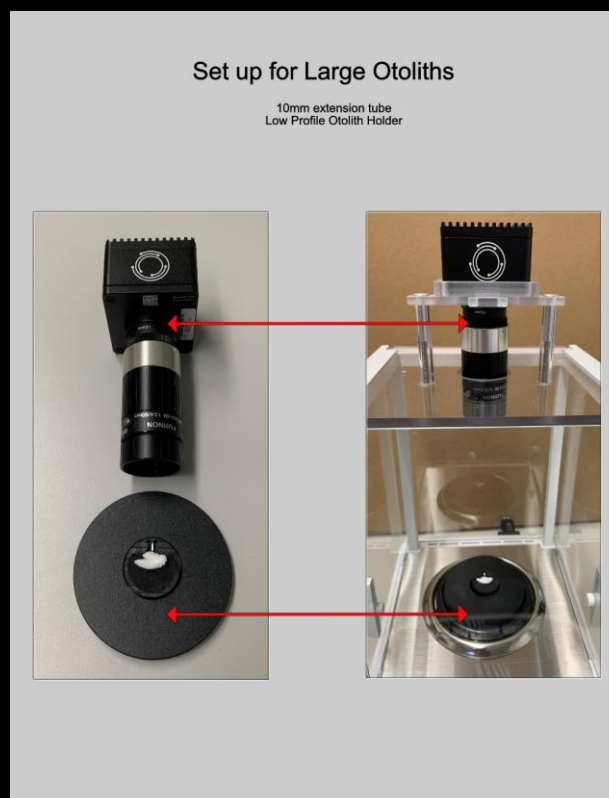
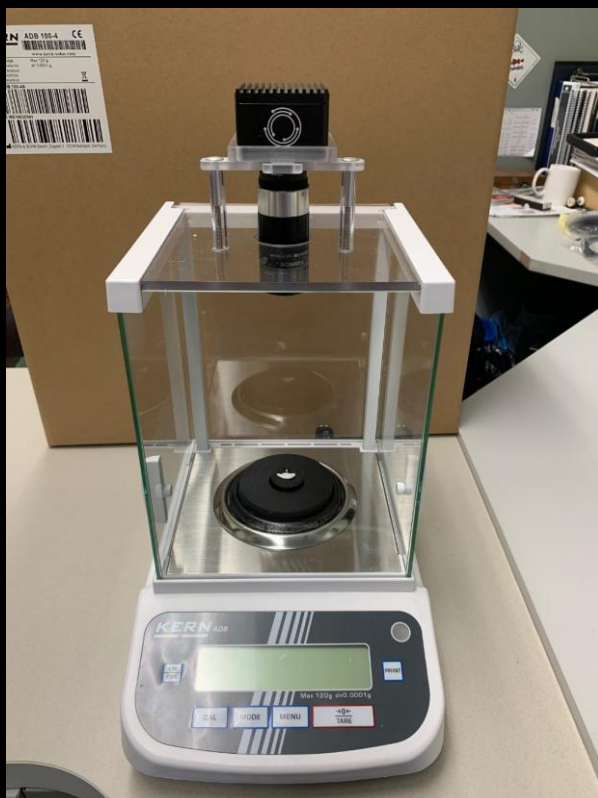
Record: 1 of 23 No Filter Search

Save Process Close

Alternative Age Estimation Equipment

Method of groundfish otolith age and attribute data collection

- Combination four decimal place balance and otolith surface image capture

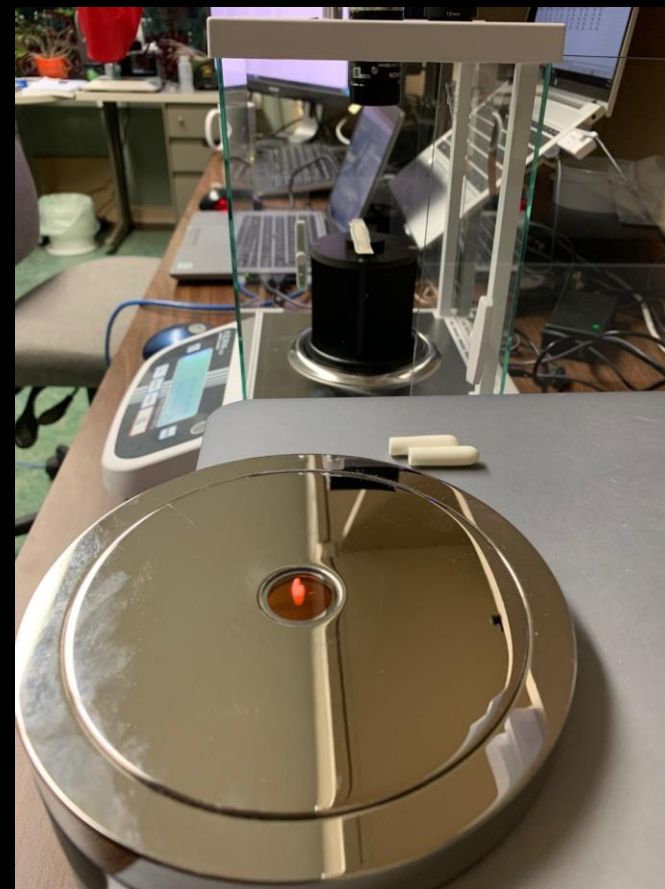
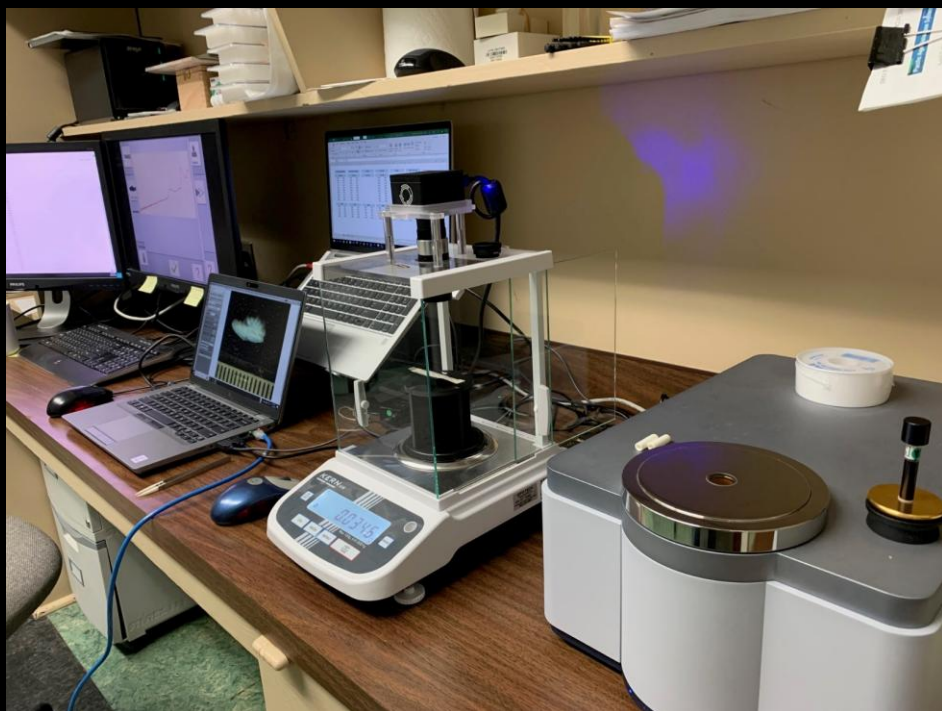


Alternative Age Estimation Equipment



Method of groundfish otolith age and attribute data collection

- Near Infrared otolith spectroscopy



Alternative Age Estimation Equipment



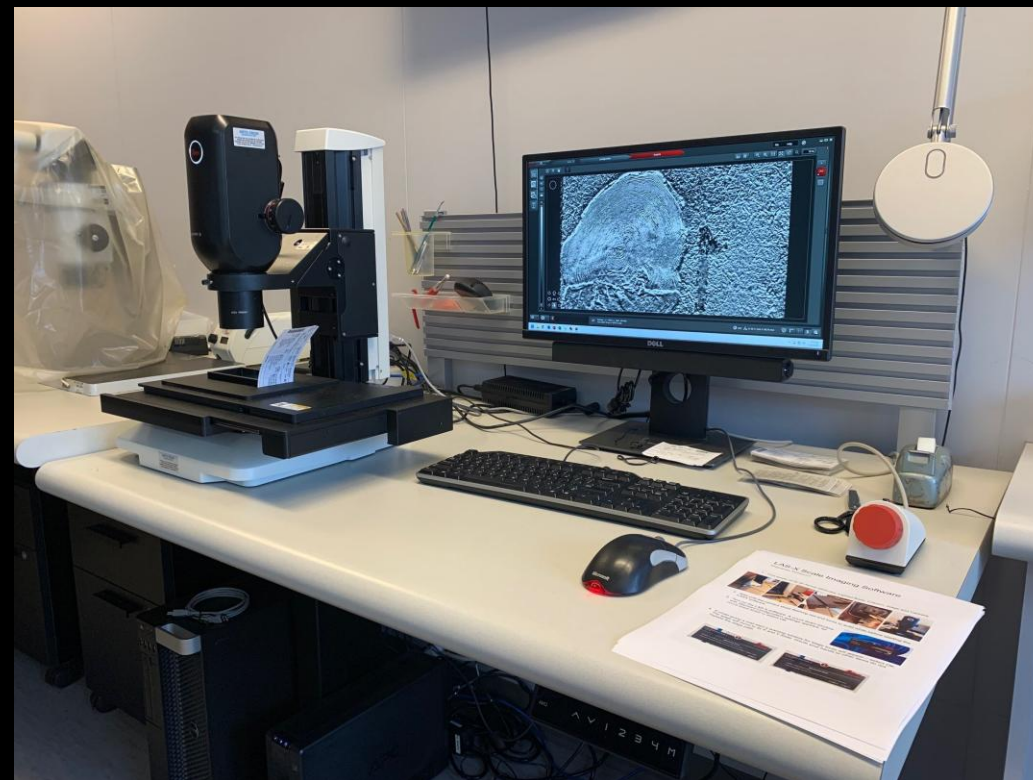
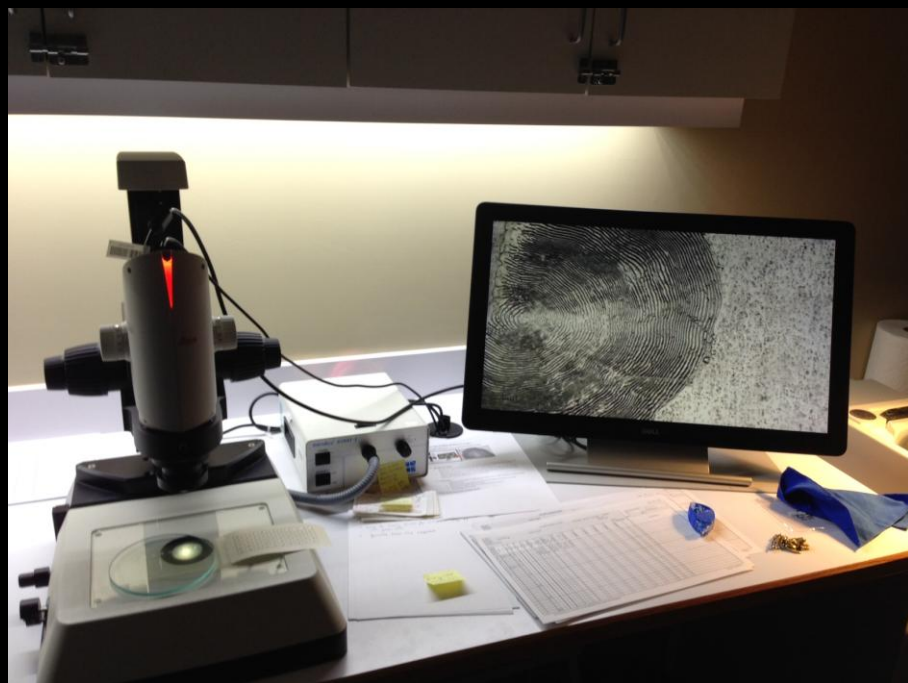
Method of groundfish otolith age and attribute data collection

- Otolith weight, imaging and IR scanning

Alternative Age Estimation Equipment

Method of salmon scale attribute data collection

- We are working on a similar SCLDDE application for salmon
- Currently we have far less data integration then we'd like
- Scale imaging



Alternative Age Estimation Equipment

Method of salmon scale attribute data collection

- Scale imaging

Alternative Age Estimation Equipment

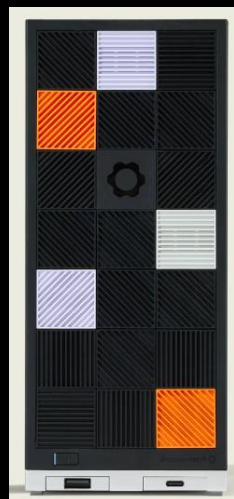
Computational Equipment

Dell Precision 7960



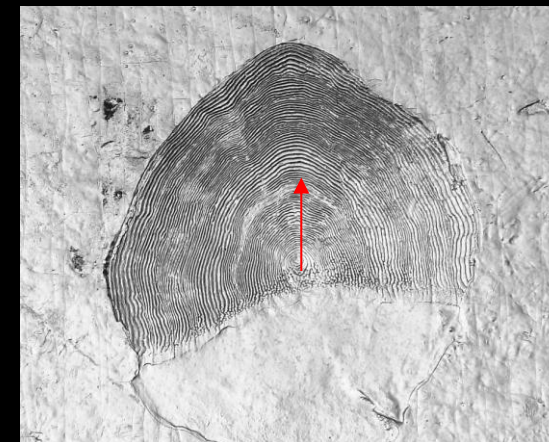
- Intel Xeon w7-3465X (28 cores)
- 128GB DDR5 4800Mhz RDIMM
- NVIDIA RTX 5000 Ada 32GB GDDR6
- 1TB M.2 PCIe NVMe Opal Encrypted SSD
- 4TB M.2 PCIe NVMe SSD
- Linux Ubuntu 22.04 operating system

Framework Desktop Computer



- Controls Leica high speed
- Creates unique specimen file names
- Extracts first marine growth measures
- Predicts Age

Chum_123_Escapement_2017-10-31_110603_1068221-2A_10sc2_41.png



Alternative Age Estimation Equipment

Scale, otolith images and IR data storage

Storage and use issues;

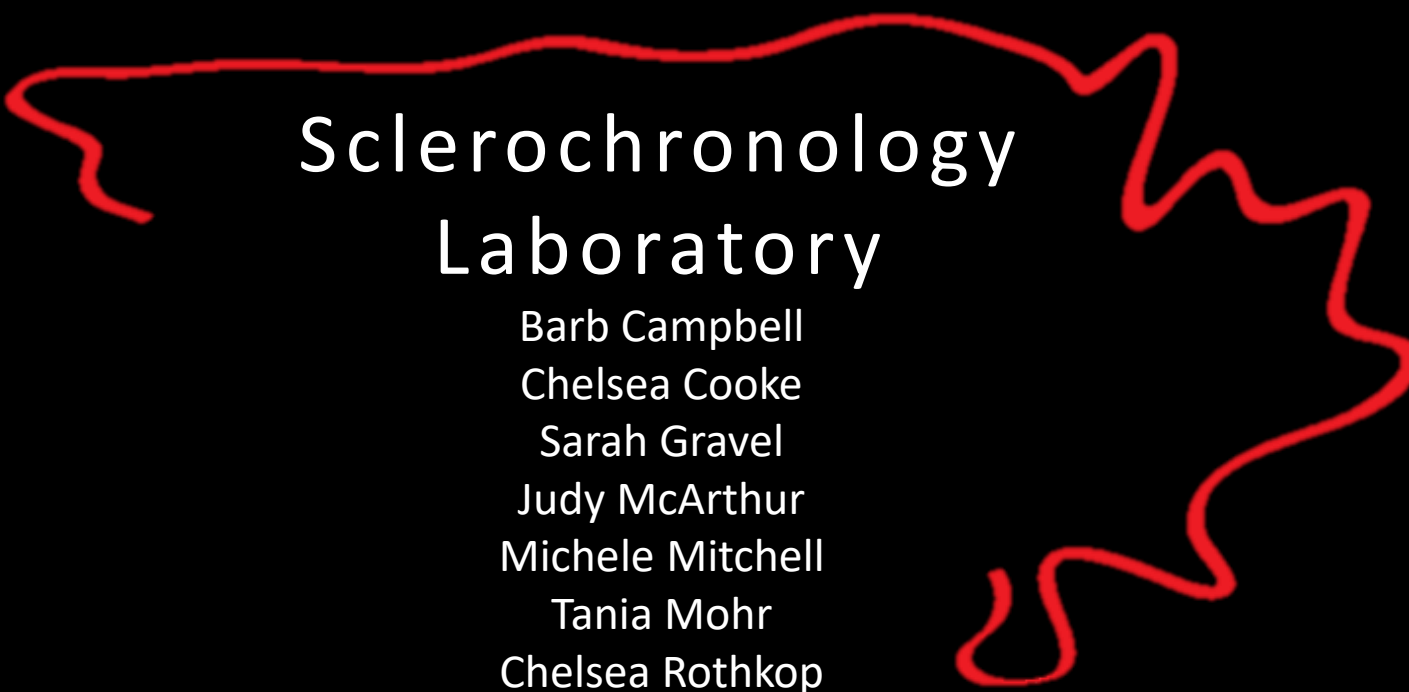
- The HDisc space required prevents local server storage
- Cloud archive storage is too expensive
- Network connectivity is too slow for algorithm cloud AI training
- Local DAS storage is our current solution
- DAS unit is RAID5 formatted for redundancy
- However, local storage in an earthquake zone where our building likely will collapse is not ideal



Alternative Age Estimation Species and Attribute Numbers

Species	Structure	Attribute	Number
Pacific Ocean perch	otolith	FT-NIR/image/wt.	3244
Rougheye\Blackspotted	otolith	FT-NIR/image/wt.	5410
Yelloweye	otolith	FT-NIR/image/wt.	3546
Quillback	otolith	FT-NIR/image/wt.	3409
Red Banded	otolith	FT-NIR/image/wt.	2400
Widow	otolith	FT-NIR/image/wt.	4000
Bocaccio	otolith	image/wt.	264
Canary	otolith	image/wt.	394
Lingcod	otolith	FT-NIR/image/wt.	169
Arrowtooth flounder	otolith	FT-NIR/image/wt.	382
Hake	otolith	image/wt.	1627
Chum	scale	Image	~12000+
Chinook	scale	Image	~8000+
Atlantic Cod*	otolith	FT-NIR	721
Eulachon	otolith	image/wt.	1169

*Collaboration Aaron Adamack Northwest Atlantic Fisheries Centre



Sclerochronology Laboratory

Barb Campbell

Chelsea Cooke

Sarah Gravel

Judy McArthur

Michele Mitchell

Tania Mohr

Chelsea Rothkop

Audrey Ty

Louisa Varco

Stephen Wischniowski

Sclerochronology Lab Research Branch

How Old Is That Fish?

ROUND FISH

Lingcod	<i>Ophiodon elongatus</i>	21
Pacific cod	<i>Gadus macrocephalus</i>	8
Pacific hake	<i>Merluccius productus</i>	22
Sablefish	<i>Anoplopoma fimbria</i>	92
Walleye Pollock	<i>Theragra chalcogramma</i>	9

FLAT FISH

Arrowtooth flounder	<i>Astheresthes stomias</i>	26
Dover sole	<i>Microstomus pacifica</i>	57
English sole	<i>Parophrys vetulus</i>	22
Flathead sole	<i>Hippoglossoides elassodon</i>	15
Petrable sole	<i>Eopsetta jordani</i>	35
Rex sole	<i>Glyptocephalus zacharus</i>	15
Rock sole	<i>Lepidopsetta bilineata</i>	25

SALMON

Chinook	<i>Oncorhynchus tshawytscha</i>	7
Sockeye	<i>Oncorhynchus nerka</i>	8
Coho	<i>Oncorhynchus kisutch</i>	4
Chum	<i>Oncorhynchus keta</i>	7
Pink	<i>Oncorhynchus gorbuscha</i>	3

HERRING

Pacific Herring	<i>Clupea pallasii</i>	15
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ROCK FISH

Blackgill Rockfish	<i>Sebastes melanostomu</i>	64
Canary rockfish	<i>Sebastes pinniger</i>	84
China rockfish	<i>Sebastes nebulosus</i>	79
Copper rockfish	<i>Sebastes caurinus</i>	51
Darkblotched rockfish	<i>Sebastes crameri</i>	48
Harlequin rockfish	<i>Sebastes variegatus</i>	48
Pacific ocean perch	<i>Sebastes alutus</i>	100
Pygmy rockfish	<i>Sebastes wilsoni</i>	24
Quillback	<i>Sebastes maliger</i>	95
Redbanded rockfish	<i>Sebastes babcocki</i>	93
Redstripe rockfish	<i>Sebastes proriger</i>	55
Rougheye rockfish	<i>Sebastes aleutianus</i>	147 (205)
Sharpchin rockfish	<i>Sebastes zacentrus</i>	58
Shortraker rockfish	<i>Sebastes borealis</i>	120
Silvergrey rockfish	<i>Sebastes brevispinis</i>	81
Tiger rockfish	<i>Sebastes nigrocinctus</i>	69
Widow rockfish	<i>Sebastes entomelas</i>	60
Yelloweye rockfish	<i>Sebastes ruberrimus</i>	128
Yellowmouth rockfish	<i>Sebastes reedi</i>	99
Yellowtail rockfish	<i>Sebastes flavidus</i>	59
Vermillion rockfish	<i>Sebastes miniatus</i>	60
Shortspine thornyhead	<i>Sebastolobus alascanus</i>	50
Longspine thornyhead	<i>Sebastolobus altivelis</i>	65